

Development and Evaluation of Collision Avoidance User Interface for Assistive Vision Impaired Navigation

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ABSTRACT

In this paper, an initial novel user interface prototype is developed and evaluated to assist vision impaired people to avoid collision with moving people while navigating in indoor environments. The user interface performs both pose classification and distance estimation based on RGB images and sends the results to the decision tree classifier model which classifies whether it is safe for vision impaired person to navigate, or should the vision impaired person be cautious or stop to avoid collision with the moving person. Experimentation showed that the user interface with the combined performance of pose classification, distance estimation and decision tree model showed an accuracy of 93.55% on a testing video dataset.

CCS CONCEPTS

• Computing methodologies → Activity recognition and understanding; Neural networks.

KEYWORDS

Vision Impaired Navigation, Pose classification, Depth estimation

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1 INTRODUCTION

Vision impaired navigation involves assistance in different navigational subtasks. One of the subtasks consists of avoiding collision with static and dynamic objects while moving which is important when the vision impaired person navigates in indoor environments, such as office rooms, warehouses, university campuses, malls etc. Works involving smartcanes and navigation aids such as [2, 8–12] focuses on avoiding collision in vision impaired navigation but they have either focused on only avoiding static obstacles or focused on implementing feedback mechanisms to make the other person move to avoid collision with visual impaired user and on only certain scenarios. Works regarding obstacle avoidance using smartphones such as [5, 6] have mostly focused on static obstacles only and

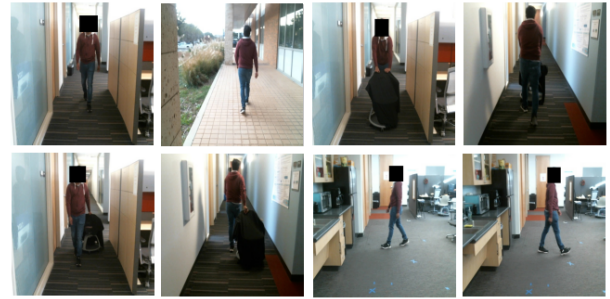


Figure 1: Poses used for pose classification (clockwise from top left): Opposite Direction, Same Direction, Push Cart Opposite, Push Cart Same, Towards Right, Towards Left, Pull Cart Same and Pull Cart Opposite

works like [3, 7] have focused on avoiding collision while standing in line which is like avoiding obstacles with little movement. Work needs to be done towards developing unique user interfaces and systems that can assist and provide path for vision impaired people in avoiding collision during navigation even while people move around in different environments. This paper focuses on the development and evaluation of an initial novel user interface prototype to assist vision impaired people in avoiding collision with moving people in indoor environments. The user interface prototype performs both vision-based pose classification and distance estimation of the person moving in front from a vision impaired person's point of view. Based on the pose and distance of the moving person, the user interface classifies whether it is safe for the vision impaired person to move or has to stop to avoid collision with the moving person. The proposed user interface prototype is novel as it assists vision impaired people to navigate in moving people scenario in indoor environments and performs pose classification along with distance estimation at the same time, for different movement poses, with just RGB images obtained from webcam and without any extra sensors such as depth sensors as employed in previous works.

2 METHODOLOGY

This paper focuses on the development and evaluation of the user interface prototype using videos of moving people as input. As shown in Figure 2, video sequence of a person moving in different poses is given as input. The user interface consists of three components - pose classification module, distance estimation module and decision tree classifier model. The pose classification module consists of a pre-trained YOLOv8 keypoint detection model [1] and a novel custom LSTM model for classifying the pose of the person moving in the video sequence. The keypoints of the person moving

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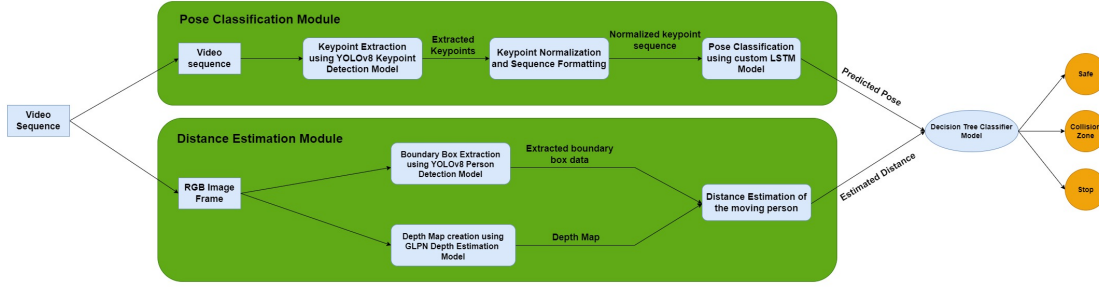


Figure 2: System Architecture of the User Interface Prototype

in each frame in the video sequence are extracted using the key-point detection model, which are normalized and formatted into sequence-based data with the sequence number and the keypoint data in each sequence. The keypoint data with the sequence detail is loaded into the custom-built LSTM model, which classifies the pose of the person moving in the video sequence. The custom LSTM model consists of a single-layer LSTM connected to a linear fully connected layer, which classifies the pose of the moving person. In this paper, the pose classification focuses on eight different poses commonly seen in indoor environments like warehouses, university campuses, malls etc. where four poses are normal movements and other four poses are replication of cart based movements. The eight poses are - Opposite Direction, Same Direction, Push Cart Opposite, Push Cart Same, Pull Cart Opposite, Pull Cart Same, Towards Left and Towards Right. Figure 1 showcases the poses, from a vision impaired user point of view, that are used for pose classification. The custom LSTM model is trained on a custom video dataset collected from 7 participants and classified into eight different poses. The participants were moving in the eight poses in front of the camera in 4 different indoor environments on the university campus. The distance estimation module consists of a pre-trained YOLOv8 based person detection model and a pre-trained depth estimation model. Each frame in the video sequence is sent to the distance estimation module. The person detection model detects the moving person and obtain the boundary box coordinates of the detected person in the image frame. The depth estimation model converts the image frame into a depth map. GLPN (Global-Local Path Networks) model trained on NYU Depth V2 dataset [4] is used as the depth estimation model. Based on the boundary box coordinates and the depth map, the distance of the moving person is estimated. The classified pose and estimated distance of the moving person is then sent to the custom decision tree classifier model which classifies the scenario into one of three classes - *Safe* (vision impaired user can safely move), *Collision Zone* (vision impaired user has to move carefully to avoid collision with the moving person) and *Stop* (vision impaired person has to stop to avoid collision with the moving person).

3 EXPERIMENTATION

The initial user interface prototype was tested in three different aspects - pose classification accuracy, decision tree classifier accuracy and combined integration test accuracy, where the complete user interface prototype was tested on a testing video dataset. The custom dataset videos collected from 7 participants consists of 473 videos,

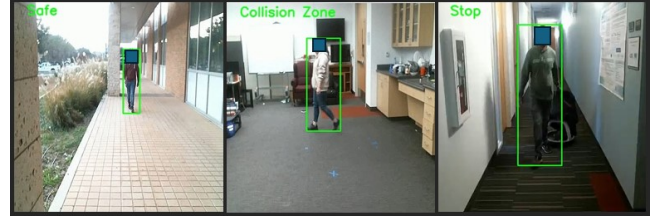


Figure 3: User interface results - *Safe* (Opposite direction), *Collision Zone* (Towards left), *Stop* (Pull Cart Opposite)

which is split into 80% training dataset and 20% testing dataset. The training dataset was used to train the custom LSTM model, and the pose classification accuracy was tested using the testing dataset. The custom LSTM model was trained for 125 epochs on the training dataset with a 0.01 learning rate. The custom decision tree classifier model was trained using the poses and distance estimated in every frame of videos in the training dataset and the accuracy was calculated using the pre-determined poses and the distance estimated using the distance estimation module in video frames from the testing dataset. The whole user interface prototype with the pose classification, distance estimation and the decision tree classifier model is then integrated and tested with the videos in the testing video dataset as input to calculate the combined integration test accuracy. The results showed that, in terms of pose classification accuracy, the custom LSTM model showed an 86% accuracy on the testing dataset. In terms of decision tree classifier accuracy, the decision tree classifier model showed an 99.85% accuracy. The combined integration test with the testing video dataset showed an accuracy of 93.55%. Figure 3 shows some of the results shown by the user interface during the combined integration test.

4 CONCLUSION AND FUTURE WORK

The initial user interface prototype shows a combined integration test accuracy of 93.55% and at the same time, work is needed to be done to develop and improve the user interface. Future work includes extending the user interface to a centralized system where multiple vision impaired people can access the system using a hardware prototype setup and receive personalized solutions to avoid collision while navigating in indoor environments. Also the LSTM based pose classification model will be trained with a large video dataset collected from more participants and the system will be tested in different indoor scenarios in real-time environments.

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